CLASS 9 · CBSE · GANITA MANJARI PART 2 · FORMULA BANK

Chapter 14: Math of Space: Surface Area and Volume

Every formula and result of the chapter on one sheet

1	Cuboid: total surface area	$2(lw + wh + lh)$
2	Cuboid: volume	lwh
3	Cube: total surface area	$6a^2$
4	Cube: volume	a^3
5	Cylinder: curved surface area	$2\pi rh$
6	Cylinder: total surface area (both ends closed)	$2\pi r(h + r)$
7	Cylinder: surface area (one end closed)	$\pi r(2h + r)$
8	Cylinder: volume	$\pi r^2 h$
9	Cone: slant height	$l = \sqrt{h^2 + r^2}$
10	Cone: curved surface area	πrl
11	Cone: total surface area	$\pi r(l + r)$
12	Cone: volume	$\frac{1}{3}\pi r^2 h$
13	Pyramid: volume	$\frac{1}{3} \times$ area of base $\times$ height
14	Sphere: surface area	$4\pi r^2$
15	Sphere: volume	$\frac{4}{3}\pi r^3$
16	Hemisphere: curved surface area	$2\pi r^2$

17	Hemisphere: total surface area	$3\pi r^2$
18	Hemisphere: volume	$\frac{2}{3}\pi r^3$
19	Lengths multiplied by k	Area $\times k^2$, volume $\times k^3$
20	Capacity	$1000\ \text{cm}^3 = 1$ litre; $1\ \text{m}^3 = 1000$ litres

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